

Characterizing Index Structures in Multi-Tenant Pipelines

Kenji Marchetti, Priya Gorecki, Caius Oyelaran, Cordelia Thibodeaux

Sableport Marine Laboratory

Section: Computing
Published: 2020-12-19
DOI: 10.5281/mjas.9.4.01
Keywords: ablation study; cohort design; cost accounting

Abstract

We examine the behaviour of index structures in edge deployments. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. The reduced model retains most of the explanatory power at a fraction of the computational cost.

1. Introduction

Sensitivity to the smoothing window proved modest across the range we examined. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses.

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2. Related Work

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3. Materials and Methods

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4. Experimental Setup

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5. Results

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6. Discussion

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Neither of the alternative formulations improved on the simple additive model in our setting. Residual error grows approximately linearly with the number of coupled parameters. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The reduced model retains most of the explanatory power at a fraction of the computational cost. Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate.

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7. Limitations

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8. Conclusion

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses.

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